THE VISUAL ATLAS OF REALITY

Wave-Substrate Theory

G−2 RESEARCH BRIEFING

The Anomalous Magnetic Moment: Landscape, Requirements, and the WST Program

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Scope of This Document

This briefing combines two research documents on the WST g−2 program: a deep research assessment of the Möbius spectral approach to lepton g−2 against the 2026 experimental and theoretical landscape, and a strategic framing document on compute-first versus lab-first evidence. Together they define what the WST program claims, what it does not claim, what the hard technical requirements are, and where the genuine open work lies.

Bottom line: Phase A — can the Möbius spectral machinery compute the coefficients? — is a mathematical/theoretical physics question. You do not need new laboratory evidence to pursue it. The bottleneck is a clean theoretical mapping, not a shortage of experimental measurements.

1. What the g−2 Problem Looks Like in 2026

The anomalous magnetic moment of a lepton is defined as:

$$a_\ell \equiv (g_\ell - 2)/2 = F_2(0)$$

where $F_2(0)$ is the Pauli form factor evaluated at zero momentum transfer. [1] This pins down exactly what any alternative computation must reproduce: not a heuristic 'magnetic self-inductance,' but the specific on-shell coefficient of the Pauli term in the electromagnetic vertex. [2]

Electron g−2: Parts-Per-Trillion Precision

For the electron, the perturbative QED series has been driven to extremely high order. The tenth-order (five-loop) contribution involves 12,672 Feynman diagrams evaluated computationally. [3] The 2023 Penning-trap measurement reports g/2 = 1.00115965218059(13) — parts-per-trillion precision. [4] Standard Model comparisons are now limited by disagreements in the fine-structure constant α itself, not by the g−2 theory. [5]

Muon g−2: The 2026 State of Play

The muon story has shifted materially. The final Fermilab measurement (2025) reports a combined precision of 127 ppb: $a_\mu(\text{exp}) = 1165920715(145) \times 10^{-12}$ (124 ppb combined world average). [7,8]

The international Muon g−2 Theory Initiative issued White Paper 2025 (WP25), which shifted the Standard Model prediction upward and reports no statistically significant tension with the experimental world average at current precision. [9] The 'muon g−2 puzzle' is now better characterized as:

- A high-precision experimental benchmark that is essentially stable
- A theory problem dominated by hadronic systematics — specifically hadronic vacuum polarization (HVP) — not a clean Standard Model vs. experiment tension

2026 framing note: WP25 reports $a_\mu(\text{SM}) = 116592033(62) \times 10^{-11}$ and a difference $a_\mu(\text{exp}) - a_\mu(\text{SM}) = 38(63) \times 10^{-11}$ — no tension at current precision. The 'puzzle' shifted from 'new physics?' to 'why do dispersive vs. lattice HVP approaches disagree?'

Implication for WST framing: describing the program as 'closing the muon gap' in 2026 will draw the immediate response: 'which gap, under which hadronic pipeline?' The WST claim needs to be scoped against this reality.

2. What Counts as Evidence — Phase A vs Phase B

The question 'do we need laboratory evidence?' splits into two fundamentally different phases:

Phase A: Can the Machinery Compute the Coefficients?

This is overwhelmingly a mathematical and theoretical physics question. To decide whether the Möbius spectral program 'works' at the level currently targeted — reproducing c_2 — you do not need new laboratory evidence. You need:

- A derivation that defines the operator/Green function unambiguously
- Specification of regularization and counterterms
- A calculation yielding the benchmark coefficient(s) from first principles [10]

The one-loop benchmark ($\alpha/(2\pi)$, Schwinger 1948) and the two-loop benchmark ($c_2 \approx -0.328479$, Petermann-Sommerfield) are both analytically fixed and historically confirmed. [6,7] The 'diagram explosion' that motivates compressed representations is real and documented — modern tenth-order treatments involve 12,672 diagrams. [8] This legitimately motivates seeking a single spectral object to replace the sum.

Phase A verdict: The bottleneck is a clean theoretical mapping from the spectral/topological object to the magnetic anomaly observable $F_2(0)$. Not a shortage of experimental measurements.

Phase B: Is This New Physics or a Reorganization of QFT?

If Phase A succeeds — if the Möbius machinery can compute c_2 — Phase B asks whether this constitutes genuinely new physics or an equivalent reformulation of standard renormalized QFT. [16]

- Equivalent reformulation: a new computational representation mathematically equivalent to renormalized QED. Still a major achievement if it reduces complexity or makes structures manifest.
- New physics / new ontology: a framework that changes predictions. Must survive many precision tests and identify where deviations occur and why they haven't been excluded.

The lab evidence already in hand is stronger than most new theories can survive. Between electron g−2 at parts-per-trillion, muon g−2 at 127 ppb, and precision α from atom interferometry (rubidium recoil: $\alpha^{-1} = 137.035999206(11)$ at 81 ppt [13]; cesium recoil: $\alpha = 1/137.035999046(27)$ [14]), any claimed deviation must explain why it hasn't already been excluded across this whole stack.

3. What WST Claims and Does Not Claim

Three Load-Bearing Claims

Three specific claims in the WST narrative are testable:

Claim 1 — Geometric recovery of the Schwinger term

WST claims a direct geometric route to a_e^(1) = α/(2π). This ties the leading Schwinger term to the ratio of two electron length scales and a 2π phase cycle. The claim is: this is not a loop diagram result but a topological consequence of the Möbius electron structure. This is Phase A level — it requires showing that the geometric object correctly isolates $F_2(0)$ at this order.

Claim 2 — Möbius spectral signature for c_2

The two-loop coefficient is:

$$A_1^{(4)} = 197/144 + \pi^2/12 + (3/4)\zeta(3) - (\pi^2/2)\ln 2 \approx -0.328478965579193...$$

WST claims the $-(\pi^2/2)\ln 2$ term has a rigorous topological origin — the Möbius twist contributes $\Delta c_2 = -\pi^2\ln 2/2 = -3.4205$, flipping the sign from the +3.09 contribution of the non-ln2 terms. [12] This is numerically correct as an observation. The nontrivial question is whether it makes sense to attribute ln2 to a Möbius twist via a specified spectral object. There is at least a plausible bridge: fermionic signs picked up on traversal of nontrivial cycles are mathematically encoded as spin structure choices and boundary conditions, and spectral objects like zeta functions are a standard language for encoding how boundary conditions change spectra. [15]

Claim 3 — Single Green function replacing diagram sums

WST proposes that a Möbius torus Green's function with odd-mode constraint should reproduce the two-loop coefficient. This resembles known reorganizations of perturbation theory (the worldline formalism of Strassler is a canonical example of diagram-free reorganization at one loop [18]), but is not a derivation until the mapping to $F_2(0)$ is explicit and renormalization-complete.

What WST Does NOT Claim

The WST documents themselves state: the 12-digit agreement for electron a_e is a triumph of standard QED, and the Möbius/topology ideas are framed as interpretive/structural rather than as an independent re-computation of the full precision value. This is the correct self-constraint. Aiming first at an independent derivation of c_2 (and then c_3) is a more coherent target than claiming 12-digit reproduction.

4. Hard Technical Requirements

To turn the WST g−2 narrative into a coefficient-level breakthrough, several items currently implicit (or absent) must be pinned down. These are structural necessities imposed by what a_ℓ is as a QFT observable.

The Minimal List

1. Precise operator identity for $F_2(0)$

The QFT definition is in terms of the vertex function and form factors. [2] Any geometric/spectral replacement must show explicitly how its output isolates the same quantity — not just numerically matches one coefficient.

2. Gauge invariance and Ward identities

The vertex is constrained by gauge invariance. The anomalous moment is tied to the finite, renormalized Pauli form factor. [2] Any spectral determinant/Green-function approach must demonstrate it respects these identities. Numerically matching one coefficient is not sufficient.

3. Renormalization prescription

Spectral sums and Green functions on compact manifolds typically contain divergences and scheme-dependent constants (zero modes, regularization constants). A Möbius Green function approach must specify exactly what is subtracted and why those subtractions correspond to standard renormalization conditions. [20]

4. Dimensionality and physical embedding

Lepton g−2 is a four-dimensional relativistic observable with spinor structure. If the framework uses a reduced internal manifold, it must show how that manifold emerges from the full dynamics and why it retains the correct tensor/spin content. [16]

5. No free parameters

Any anisotropy parameter ε in the Green-function expression would be fatal unless ε is derived from independently fixed geometry — tied to α, mass ratios, or another already-fixed parameter. Otherwise, reproducing a single coefficient could be curve-fitting in disguise.

Key constraint: WST has no free ε. The coefficients c_2 and c_3 are derived from the fixed Gram matrix H and scheme constant $B_0(H)$. This satisfies requirement 5 — but requirements 1–4 remain to be demonstrated.

5. A Concrete Path Forward

Reinterpret the WST Sum as a Restricted Lattice Green Function

The WST spectral object is in the family of torus lattice sums closely related to Epstein zeta functions. A concrete research-grade route to decide whether it can produce $A_1^{(4)}$:

- Define the regulated object G_M precisely — including what happens at (m,n) = (0,0) and what is removed [20]
- Convert the lattice sum into theta-function / modular form expressions via Poisson resummation
- Take the relevant limit (coincident-point limit or the integral functional of G_M that WST asserts corresponds to c_2)
- Extract constants and compare to 197/144, $\pi^2/12$, $\zeta(3)$, and $-(\pi^2/2)\ln 2$ [21]

If the restricted 'odd-n' condition genuinely acts like a Möbius twist, it should appear as a switch between theta-characteristics or as an explicit factor producing ln2-type constants in the determinant/Green-function normalization.

Use Zeta-Regularized Determinants

If the WST storyline is 'the twist changes the spectrum; the spectrum shift yields ln2,' the correct formal language is: define a spectral zeta function ζ_L(s) for a relevant elliptic operator L, and relate the determinant to ζ_L′(0). The physically relevant quantities are differences of log determinants between twisted vs untwisted boundary conditions, which can isolate exact constants like ln2. [22]

Align with the Non-Diagrammatic Derivation Tradition

The historical non-diagrammatic route (Sommerfield: compute self-energy in a constant magnetic field; extract the term linear in B to get the magnetic moment correction [17]) is closer to the WST back-reaction framing than popular 'vertex correction' stories. A modern analogue is the worldline approach (Strassler [18]): one-loop effective actions as a one-dimensional path integral that resums diagram classes while preserving gauge-invariant structure.

A realistic WST breakthrough path: show that the WST Green-function object is the kernel of an effective action/self-energy operator in an external magnetic field (Sommerfield-style), then show that when expanded to order α^2 it reproduces the analytic $A_1^{(4)}$ constant combination. [23]

Falsification-Grade Targets

A breakthrough claim becomes compelling when it makes a prediction not already built into definitions or adjustable knobs:

- Derive (not fit) $A_1^{(4)} \approx -0.3284789656$ from the Möbius Green-function program, including the exact rational/transcendental decomposition [12]
- Extend to $A_1^{(6)}$ (three-loop, analytically known) to show the method is not tuned just to reproduce one coefficient
- Show that the same spectral object, without modification, gives the correct coefficient at each loop order — this is the hardest but most decisive test

6. The Muon Side: A Realistic 2026 Framing

The WST narrative frames a 'master breakthrough' as closing the remaining gap for the muon. Against the 2026 landscape, this requires sharper framing.

What Would Count as a Real Muon Contribution

The WSO contribution to the muon anomaly would need to engage hadronic structure directly. Muon g−2 theory uncertainty is dominated by hadronic contributions — HVP primarily, then HLbL. [6] The immediate quantitative burden:

- Can WST reproduce (or predict) hadronic contributions with uncertainties competitive with lattice-QCD methods? [26]
- Can it explain why different experimental inputs in the dispersive approach produce inconsistencies significant enough that WP25 leaned on lattice averages? [25]

If WST cannot engage these, then 'harmonic scaling' remains an interpretive metaphor rather than a calculational program for the muon anomaly.

Where the Real Scientific Leverage Is

The current situation is:

- Experiment: essentially finalized at Fermilab, world-leading, 127 ppb [7,21]
- Theory: dominated by hadronic systematics, not a clean new-physics question
- The tension moved from 'standard model vs experiment' to 'dispersive vs lattice HVP approaches'

For WST, the cleanest 2026 claim is: Phase A on c_2 for the electron. That's a well-defined mathematical target with no experimental ambiguity, a fixed benchmark, and no hadronic noise. If the Möbius spectral program can derive $c_2 = -0.328479...$ from first principles, that is a real result — independent of the current muon situation.

Strategic recommendation: Focus Phase A on the electron c_2 derivation. It is the cleanest falsifiable target: one number, no hadronic inputs, fixed benchmark, no parameter freedom. If the method works there, the muon follows as an extension.

References

Sources from both research documents, consolidated and deduplicated.

[1] Definition of a_ℓ ≡ $F_2(0)$: standard QFT vertex function formalism. Theory of the muon g−2 — Nyffeler, KPH (2014). iannounce.physik.uni-mainz.de

[2] Gauge invariance and Ward identity constraints on the electromagnetic vertex.

[3] Anomaly approximated — RIKEN. riken.jp/en/news_pubs/research_news/rr/7157

[4] Measurement of the Electron Magnetic Moment — Phys. Rev. Lett. 130, 071801 (2023). journals.aps.org/prl/abstract/10.1103/PhysRevLett.130.071801

[5] Standard Model g−2 comparison limited by α uncertainty: from [4] above.

[6] Hadronic vacuum polarization and HLbL as dominant theory uncertainty for muon: The anomalous magnetic moment of the muon in the Standard Model — Illinois Experts. experts.illinois.edu

[7] Fermilab final muon g−2 result 2025: Accepted paper — Phys. Rev. Lett. journals.aps.org/prl/accepted/be071Y40S051439596f884b0013e7ac41f07db37f

[8] Same as [7] — combined Fermilab + world average.

[9] White Paper 2025 (WP25): The anomalous magnetic moment of the muon in the Standard Model: an update — OSTI.GOV. osti.gov/biblio/2569225

[10] Phase A/Phase B framing: theoretical mapping is the bottleneck, not experimental data.

[11] Schwinger 1948 one-loop result α/(2π): On Quantum-Electrodynamics and the Magnetic Moment of the Electron — Phys. Rev. 73, 416 (1948). journals.aps.org/pr/abstract/10.1103/PhysRev.73.416

[12] Two-loop coefficient $A_1^{(4)} \approx -0.328478965...$: Quantum electrodynamics calculation of lepton anomalous magnetic moments — Progress of Theoretical and Experimental Physics (2012). academic.oup.com/ptep/article-abstract/2012/1/01A107/1569493

[13] Rubidium recoil α measurement at 81 ppt: Determination of the fine-structure constant with an accuracy of 81 parts per trillion — Nature. nature.com/articles/s41586-020-2964-7

[14] Cesium recoil α measurement: Measurement of the fine-structure constant as a test of the Standard Model — PubMed. pubmed.ncbi.nlm.nih.gov/29650669

[15] Spin structure and boundary conditions in spectral geometry: Annales de l'Institut Henri Poincaré. math.mit.edu/~sbenoist/Doc/LacetSLE2.pdf

[16] 4D relativistic structure of g−2 as a QFT observable with spinor content.

[17] Sommerfield non-diagrammatic derivation via self-energy in constant magnetic field: The magnetic moment of the electron — ScienceDirect. sciencedirect.com/science/article/pii/0003491658900034

[18] Worldline formalism — Strassler: SLAC-PUB-5757 (1992). slac.stanford.edu/pubs/slacpubs/5750/slac-pub-5757.pdf

[19] Explicit operator identity requirement for $F_2(0)$ in geometric frameworks.

[20] Torus Green function zero-mode and prime sum: Fourier series of modular graph functions — ScienceDirect. sciencedirect.com/science/article/am/pii/S0022314X18301318

[21] Analytical target $A_1^{(4)}$ rational/transcendental decomposition: from [12].

[22] Zeta-regularized determinants and det_ζ(L) = exp(−ζ_L'(0)): standard spectral geometry.

[23] Sommerfield-style effective action route for WST spectral object.

[24] Fermilab result as benchmark for years to come: Fermilab Muon g−2. muon-g-2.fnal.gov

[25] WP25 no-tension result and hadronic pipeline contingency: from [9].

[26] Hadronic contributions competitive with lattice-QCD: from [6,9].

[27] J-PARC independent muon measurement: J-PARC Project Newsletter No.99, July 2025. j-parc.jp/c/en/topics/2025/07/23001590.html

[28] PHYSICAL REVIEW LETTERS 135, 101802 (2025) — latest Fermilab result. journals.aps.org/prl/pdf/10.1103/7clf-sm2v

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